

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Expt 6</b> | <b>Finite Word length Effects</b>                                                                                 |
| <b>Date:</b>  | <b>6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB</b> |

**Code:**

```

clc;
clear;
close all;

%% Signal
fs = 1000;      % Sampling frequency (Hz)
f = 50;         % Signal frequency (Hz)
t = 0:1/fs:0.1; % Time vector

x = sin(2*pi*f*t); % Original signal

%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};

figure;

for i = 1:length(L)

    % Uniform Quantization
    xq{i} = round(((x+1)*(L(i)-1)/2))*(2/(L(i)-1)) - 1;

    % Quantization Error
    e{i} = x - xq{i};

    % Mean Square Error
    mse(i) = mean(e{i}.^2);

    % Plot Quantized Signal
    subplot(4,1,i+1);
    stairs(t,xq{i},colors{i},'LineWidth',1.2);
    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
    xlabel('Time (s)');
    ylabel('Amplitude');
    grid on;
end

%% Original Signal
subplot(4,1,1);
plot(t,x,'b','LineWidth',1.5);

```

```

title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% Display MSE
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));

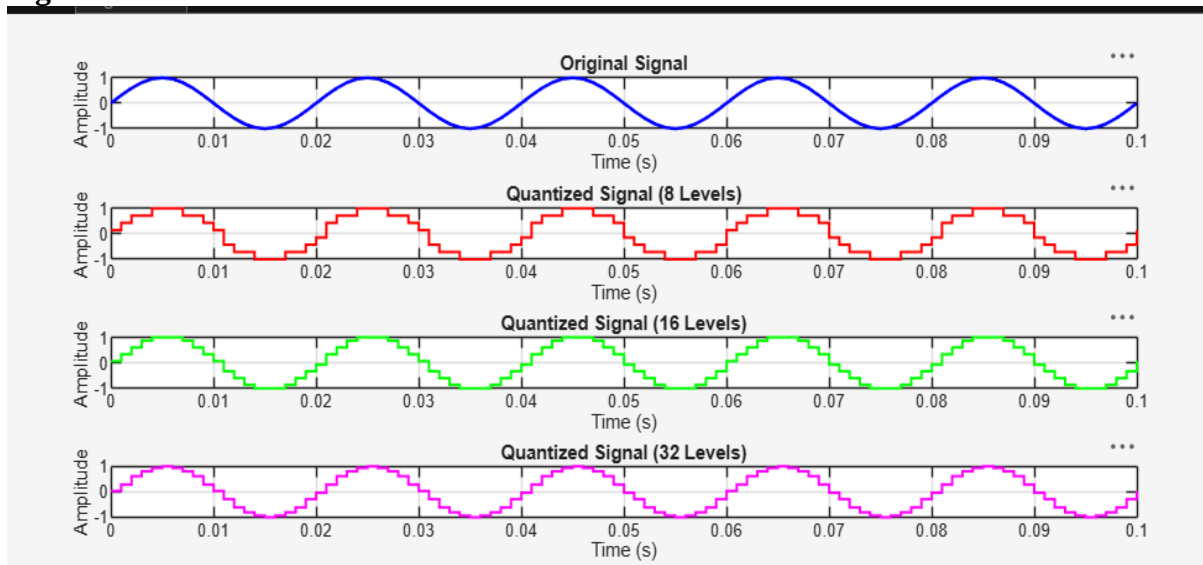
%% Error Plots
figure;

for i = 1:3
    subplot(3,1,i);
    plot(t,e{i},'LineWidth',1.2);
    title(['Quantization Error ( ' num2str(L(i)) ' Levels)']);
    xlabel('Time (s)');
    ylabel('Error');
    grid on;
end

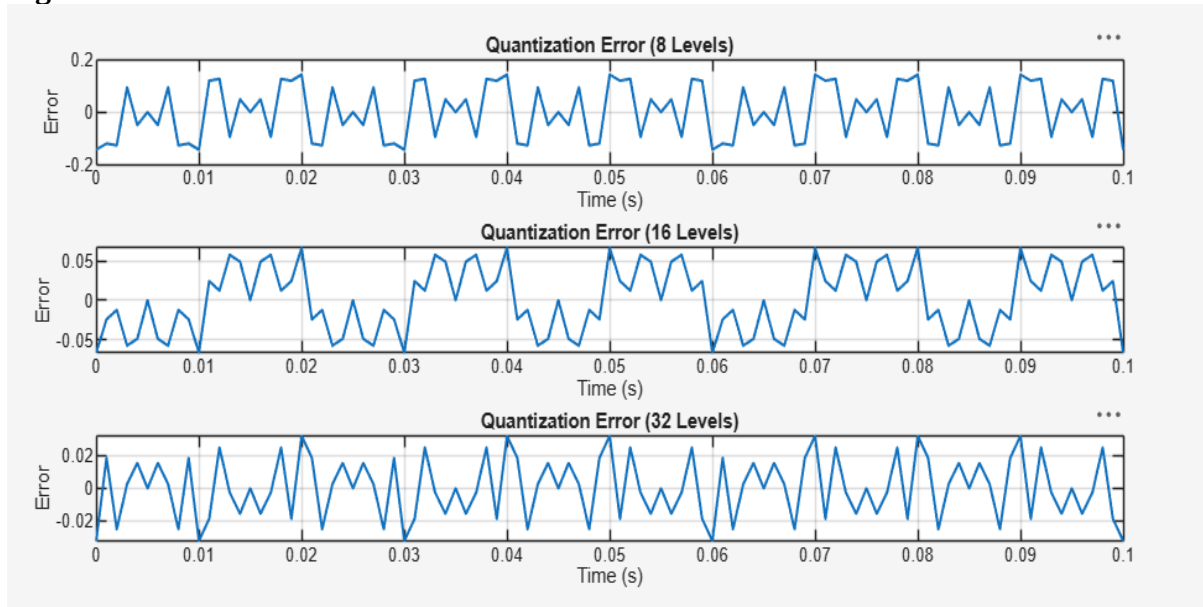
```

**Output:**

**Fig 1:**



**Fig 2:**



|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Expt 6</b> | <b>Finite Word length Effects</b>                                                                                                     |
| <b>Date:</b>  | <b>6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB</b> |

**Code:**

```
clc;
clear;
close all;

%% Generate Input Signal
n = 0:79;           % Sample index
x = sin(2*pi*n/40); % Original sine wave

%% Quantization Parameters
q = 8;              % Quantization factor

%% Quantization
xt = floor(x*q)/q;   % Truncation Quantization
xr = round(x*q)/q;   % Rounding Quantization

%% Quantization Errors
et = x - xt;         % Truncation error
er = x - xr;         % Rounding error

%% Mean Square Error (MSE)
mse_trunc = mean(et.^2);
mse_round = mean(er.^2);
```

```
fprintf('MSE (Truncation) = %f\n', mse_trunc);  
fprintf('MSE (Rounding) = %f\n', mse_round);
```

%% Figure 1: Original vs Rounded Signal

```
figure;  
plot(n,x,'b','LineWidth',1.5);  
hold on;  
plot(n,xr,'r','LineWidth',1.5);  
grid on;  
title('Original vs Quantized Signal');  
xlabel('Sample Number');  
ylabel('Amplitude');  
legend('Original','Quantized');
```

%% Figure 2: Truncation vs Rounding

```
figure;  
plot(n,x,'b','LineWidth',1.5);  
hold on;  
plot(n,xt,'r','LineWidth',1.5);  
plot(n,xr,'g','LineWidth',1.5);  
grid on;  
title('Truncation vs Rounding');  
xlabel('Sample Number');  
ylabel('Amplitude');  
legend('Original','Truncated','Rounded');
```

%% Figure 3: Quantization Error

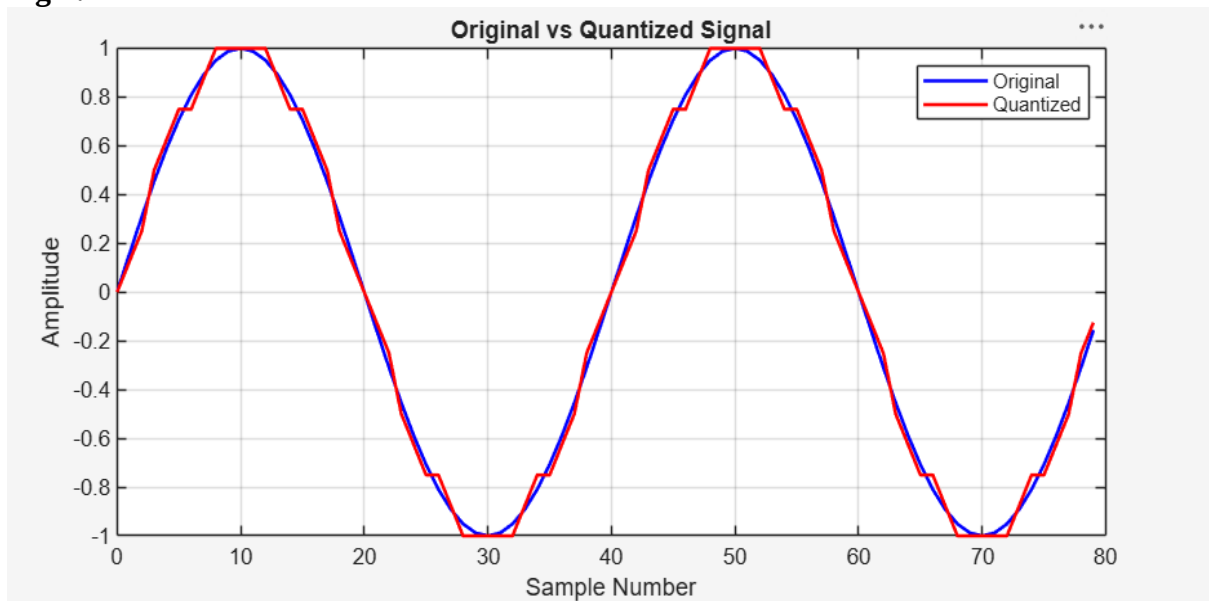
```
figure;  
stem(n,et,'r','filled');  
hold on;  
stem(n,er,'g','filled');  
grid on;  
title('Quantization Error');  
xlabel('Sample Number');  
ylabel('Error');  
legend('Truncation','Rounding');
```

%% Figure 4: Mean Square Error Comparison

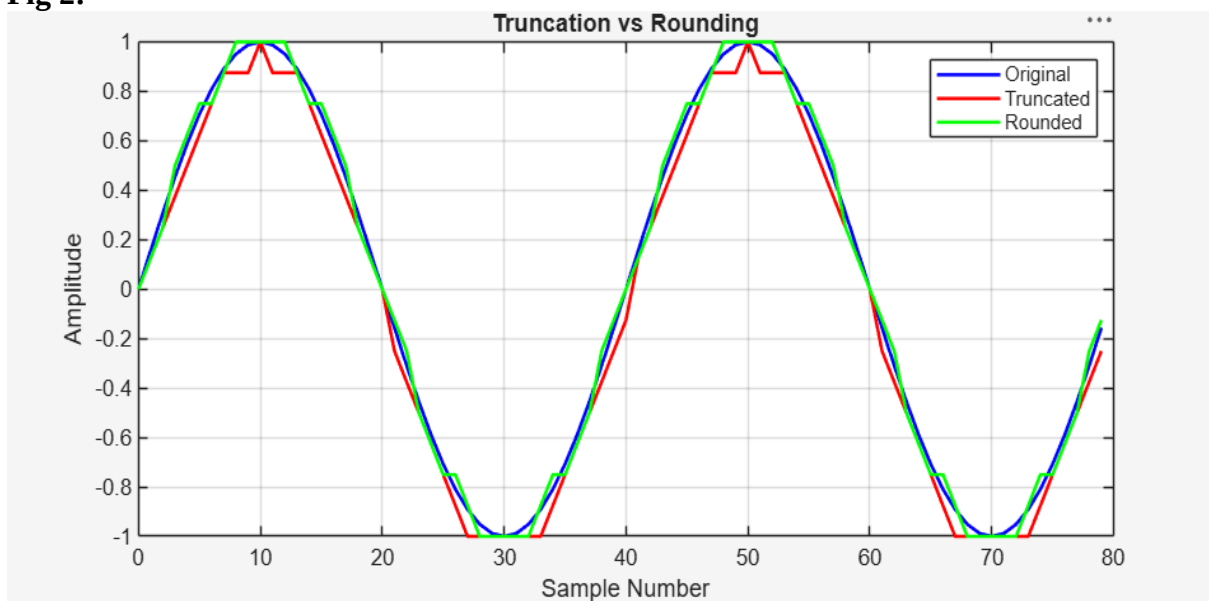
```
figure;  
bar([mse_trunc mse_round]);  
set(gca,'XTickLabel',{'Truncation','Rounding'});  
grid on;  
title('Mean Square Error (MSE) Comparison');  
ylabel('MSE');
```

**Output:**

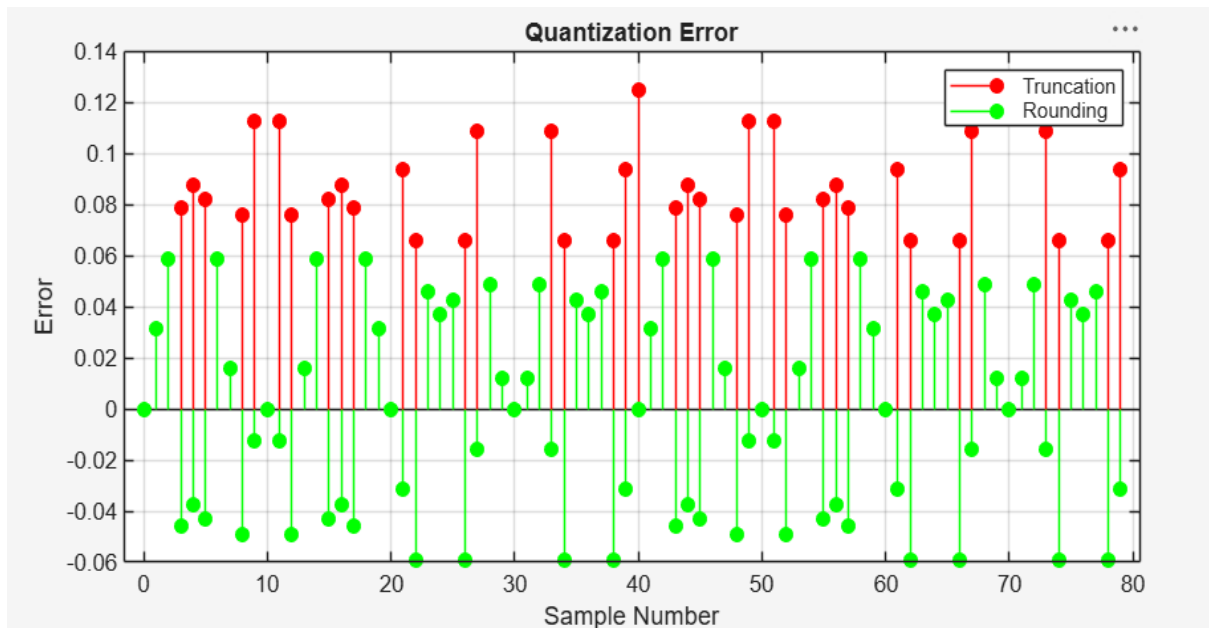
**Fig 1:**



**Fig 2:**



**Fig 3:**



**Fig 4:**

